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MANAGING TO INNOVATE IN HIGHER EDUCATION

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ABSTRACT: This paper reviews and discusses the nature of innovation in higher education teaching and learning. It traces a gradual shift from innovation generated predominantly at the local level to a form of innovation largely directed by the higher education institutions. It argues that the study of innovation demands that questions are asked about the nature and ownership of the innovation, its policy context and whose interests the innovation serves.

Keywords: innovation, teaching, learning, higher education

1. INTRODUCTION

Understandings of 'innovation' depend on where and why it occurs in institutions, organisations and systems. In higher education the 'innovator' may be a person, a team or a committee, or a government department or funding agency. When the Dearing report on higher education recommended 'to stimulate innovation' as one function of its proposed Institute for Learning and Teaching (Dearing, 1997, p. 128) it thereby added further policy clothes to an already extensive wardrobe of policy-driven innovation. The history of innovation since the Second World War has also involved the use of educational technology; individual experimentation; curriculum innovation; responses to changing student numbers, structures and funding; problem-based, resource-based, open and other strategies of learning. The story is one of significantly changing meanings.

People innovate, institutions innovate and manage innovation, and national funding agencies sponsor innovation, but the intentions and meanings differ. At system and institution levels innovation is defined explicitly, if often ambiguously within frameworks, priorities, and directions. It is decreasingly possible in higher education to innovate outside these frameworks, since funding, reputation, and even professional survival may depend on working within them. For an individual to pursue innovation outside strategic planning and responses to internal and external review and judgment may be seen at best as eccentric, at worst as dangerous – which is the range of perceptions of their work suggested by a large number of innovators to a research project on the experience of innovation.[1] Nevertheless, people do innovate, for all sorts of reasons, sometimes even with encouragement, recognition, or reward.

The Dearing emphasis on stimulating innovation in learning and teaching by-passes, at this point in its argument, other kinds of innovation with which we are also less concerned here. Curriculum innovation within departmental, faculty, and institutional frameworks is often (unlike individual innovation in teaching and learning) the product of formal, policy-related decision making, or dependent upon it. Institutional innovation has included such areas as management and governance as well as curricular and learning strategies. In the United States, for example, innovation has most frequently been concerned with institutional policies and strategies. In relation to learning and teaching there are often reservations about the value of a vocabulary of 'innovation' rather than, say, 'development', which has less of a connotation of novelty and more one of serious and safe planning. 'Innovation' does, however, have currency in higher education, though a more established one in industrial production and related processes.

2. Into the 1970s

The use in higher education of 'innovations' (in the plural) as a description of outcomes, 'innovation' as a description of a process, and the 'innovator' as a category of actor was a salient feature particularly of the 1970s. Although programmed learning, the use of 'audio-visual aids' and 'educational technology', and experiments with teaching strategies of various kinds marked earlier decades of the century, the aggregated impact on higher education was extremely limited. Until the mid-1970s innovation in British higher education was embodied in three main, discrete forms. *First,* from the beginning of the 1960s, the new 'green fields' universities were exploring new curricular structures, including 'area studies' and other interdisciplinary programmes, and the Open University emerged as a major innovation and force for innovation in higher education, making its first teaching broadcast in 1973. *Second,* the post-war momentum of 'a.v.a.' and 'ed tech' moved from programmed learning ('instruction' in the US) to tape-slide formats, overhead projects, video, closed-circuit television, and other exploitations of technologies that were becoming available on the market. Some of these forms of innovation related primarily to schools and teacher education, others to a small number of enthusiasts in higher education. *Third,* from the 1950s the work of Jane Abercrombie in particular had begun to stimulate interest in small-group teaching. Innovative practices in teaching and learning in higher education did not spread, however, beyond limited pockets of interest. The Hale committee on *University Teaching Methods* in 1964, found a very patchy use of audio-visual aids, 'a small minority' of staff using sound-recorded material 'a few' references to closed-circuit television, and two universities where research was taking place on the possible use of programmed learning. At 'most' universities in 'most' subjects the lecture remained 'the main vehicle for instruction' (Committee on University Teaching Methods, 1964, pp. 52, 98–102).

The emergent interest of the 1970s was not just in the scatter of curricular, media and teaching 'innovations', but also in constructing cumulative or strategic approaches to 'innovation', especially in teaching and learning. The Nuffield Foundation established a Group for Research and Innovation in Higher Education in 1972 and it reported over a four-year period on a modest range of course developments and teaching activities in the universities and in two-thirds of the recently created polytechnics. The teaching developments reported by the Group were from the universities, and although the polytechnics and the Council for National Academic Awards were taking various kinds of initiatives there was little innovation specifically in learning and teaching. The polytechnics were preoccupied with institutional restructuring, but also with such new curricula as business studies and the possibilities of modular structures. There were, of course, pedagogical implications of such processes as the conduct of sandwich courses, which had been established by the CNAA's predecessor, the National Council for Technological Awards. The Nuffield Group found that it was in some disciplinary areas of the universities (a significant proportion of its examples were in engineering and science) that teaching innovation was mainly to be found.

The Nuffield Group was operating at a time of deteriorating national economics in the early 1970s, and innovation ceased to be identified with the initiatives of newly created institutions, except for those of the Open University. The Group's first newsletter in 1973 reported on 'numerous' changes in curricula and 'the development of new teaching/learning situations', some of the latter involving 'more active participation in the learning process' (Nuffield, 1973a, pp. 3, 9). The investigation reflected and encouraged an interest in innovation at a time of severe cutbacks: 'the climate has changed. ... Our own project came right on the cusp of this change' (Nuffield Group in *THES*, 1976, p. 9). Other examples of investigation and debate were emerging, sometimes involving researchers or developers who were connected with the Nuffield Group in various ways. The Society for Research into Higher Education had come into existence in 1965–6, shortly after the Robbins report on *Higher Education*, and from then on it became a vehicle for discussion of both innovations and innovation strategies. For example, in 1966 it brought together a group of members with an interest in studying methods of teaching in higher education (Abercrombie, 1970, Foreword). In 1969 it set up a Study-Group on Innovation in Higher Education, convened by K. G. Collier, Principal of Bede College, Durham, and including: Jane Abercrombie, then head of an Architectural Education Research Unit at University College London; Ruth Beard, formerly head of the University Teaching Methods Unit at London University Institute of Education and at this time Professor in a Postgraduate School of Studies in Research in Education at the University of Bradford; and Tony Becher, Assistant Director of the Nuffield Foundation and director of the Nuffield Group.

These all helped to shape the new interest in teaching and learning. In 1974 Collier edited *Innovation in Higher Education*, which contained a chapter by Abercrombie on 'improving the education of architects' and one by Beard on 'promoting innovation in university teaching', together with other case studies of teaching projects and innovations (Collier, 1974). In 1970 Penguin Books had published Beard's *Teaching and Learning in Higher Education*, which covered theories of learning and accounts of programmed learning, computers, independent study and a range of new techniques, backed up by a bibliography in which medical and science education featured prominently (Beard, 1970). Her *Research into Teaching Methods in Higher Education* went through four editions (published by SRHE, with the addition of co-authors) between 1967 and 1978, responding increasingly to research associated with 'the efficiency and productivity of university education' (Beard, 1978, p. 14). Abercrombie was a member of what became the SRHE's Working Party on Teaching Methods in Higher Education, and was continuing her pioneering work in group teaching. For SRHE she wrote *Aims and Techniques of Group Teaching*, which went through four editions between 1970 and 1979 (Abercrombie, 1970). By the fourth edition, but also in Abercrombie and Terry *Talking to Learn: Improving Teaching and Learning in Small Groups* (Abercrombie and Terry, 1978), she was reporting on projects funded by the University

Grants Committee from 1972. The work to develop small-group teaching had been advocated by the UGC in 1964 and the National Union of Students in 1969 (Ibid., p. 1). From the third edition of *Aims and Techniques* she was suggesting that 'the struggle for recognition of the claims of teaching in small groups had already succeeded, at least in many places' (1979, Preface).

Among other activities and publications, in 1971 SRHE held its annual conference on the theme of 'Innovation in higher education' (Flood Page and Greenaway, 1971) and in 1973 a European symposium, the proceedings of which were published as *Individualising Student Learning: The Role of the New Media* (SRHE, 1974). The first session was chaired by Ruth Beard, and a key contribution was made by Professor Lewis Elton of Surrey University. Elton was a key player in the field, not only taking part in developments at Surrey and in science education in higher education generally, but also chairing a CNAA committee on teaching methods from 1975 (Council for National Academic Awards, 1981, p. 84). He was to write extensively on innovations in the field. In this 1973 symposium Elton outlined and analysed the 'upsurge' from the early 1960s of an interest in new teaching methods, including individualised self-paced work, and he emphasised higher education's debt to the Nuffield Foundation for funding and investigating innovations (SRHE, 1973, pp. 2–6). The Nuffield Group reviewed Elton's work to introduce the Keller Plan on independent learning at Surrey and he took part in the Group's conference on 'independence in learning' (Nuffield, 1973b, p. 20; 1974, p. 20).

3. Individual, Guided and Directed

Innovation was now much more firmly on the higher education agenda. Description and discussion appeared in the pages of *Universities Quarterly* and the SRHE's own *Studies in Higher Education.* Journals of educational media development had begun to proliferate in the US and Britain in the 1960s, under the particular impetus of programmed learning. An International Council for Educational Film changed its name in 1966 to the International Council for Educational Media (with a secretariat in Paris). The first issue of its *Educational Media International* appeared in 1976, published in London, and it reported on discussion in an 'Innovation and Information Working Group' International Council for Educational Media (1978). The literature of independent, flexible, open, and other forms of learning reflected the interests of psychologists, the exigencies of the classroom and the pressures of policy, economics and new priorities, including a focus on teacher–student interaction, the impact of competing directions within a diversifying higher education system, and the availability of new technologies and media. Confidence in conventional lecturing as a means of stimulating thought rather than simply presenting information was being undermined (Bligh, 1971).

Into the late 1970s the innovator was still largely the individual enthusiast, perhaps tuning in to opportunities opened up elsewhere, adapting new media, occasionally responding to the international outputs of the psychology of learning, or simply experimenting with something different. From the late 1970s the contents, practice, and understanding of innovation changed. The most directly influential impact was that of institutional structural changes, themselves responding to systemic change. The context had become one of accountability, the thrust of which was determined by Labour policies and made pre-eminent by the Thatcher administration from 1979. The increased public visibility of higher education practices in the late 1970s and 1980s turned powerful searchlights on all aspects of higher education operations in the university and polytechnic sectors (which became a single sector in 1992), including the quality of teaching. Sectors and institutions responded differently, but existing structures for staff and educational development were strengthened, or new ones created. Efforts at innovation did not diminish across the period, but its functions were increasingly institutionalised. A period of 'guided innovation' had begun.

An increasing policy and funding emphasis on improvements in teaching coincided with political attacks on 'incompetent teachers' in higher education. The Government's 1985 Green Paper expressed as one of its 'main concerns' the issue of 'quality of teaching', and focused the question of quality 'first and foremost on the quality, competence and attitudes of academic staff' (Secretaries of State, 1985, pp. 4, 28). Staff appraisal entered the agenda. An additional pressure from the 1970s was that of expansion: 'Hitherto, universities had given little thought to the way teaching is carried on or to ways of measuring its effectiveness. It has required the duress of growth to make them do so' (Mackenzie *et al.*, 1976, p. 18). Underpinning policy shifts, however, was also the increasing realisation that institutions were not only accountable but also in need of more coherent approaches to the use of computers and other new technologies as aids to learning. In the mid-1960s when the Brynmor Jones committee reported on *Audio-Visual Aids in Higher Scientific Education* (Jones, 1965) educational computing was only just beginning its journey into higher education, but by the end of the 1970s it was one of the driving forces of policy making on teaching and learning nationally and institutionally. One educational technology activist reflected in 1985 on the changes of the previous two decades:

> We no longer worship at the altar of 16mm films, or linear teaching machines, or multi-media approaches, or feedback classrooms: now it is the computer and its flamboyant half-brother, interactive video. There is one notable difference: computers are glamorous, up-market and, best of all, superb generators of money (Unwin, 1985, p. 65).

'Guided innovation' in teaching and learning was destined to be increasingly subordinated to 'guided new technology development' and what Americans called 'improvement of instruction'. Innovation had become strategic, and governments were involved in promoting innovation for a cluster of strategic purposes. The lead was set by the United States with considerable federal and other funds being invested in the promotion of improved learning. From the mid-1960s the US Department of Education and other agencies used a variety of strategies to promote improvement, and invested strongly in doing so (Hawkridge, 1978, pp. 61–4). Between 1966 and 1969 the American government alone invested more than $2.5 billion in experimentation and research with 'instructional technology' (Carnegie, 1972, p. 45 citing Molnar). A stream of directories and analyses emanated from government and private sources listing and explaining resources and methods concerning 'technologies for learning' (e.g. Weisgerber, 1968; Meisler, 1971). A typical directory of *New Media and College Teaching* in 1968 covered everything from television, films and telephone to programmed instruction, computer-aided instruction, overhead projectors, and simulation. It concluded with messianic confidence that all these 'breakthroughs toward improved instruction are opened by the waves of experimentation that are inundating campuses of all sorts in all parts of the nation', and resistance based on 'inertia and poverty' was being eroded by 'the increasing force of successful examples'. The future 'for anyone who believes that new media can encourage better teaching is bright' (Thornton and Brown, 1968, p. 46). In Britain the same trend was apparent, including under the stimulus of the Hale Committee in 1964 (Committee on University Teaching Methods, 1964; Unwin, 1969). The process accelerated in the 1970s and 1980s as the economic imperatives grew stronger and higher education faced increasingly difficult problems.

In Britain the 'waves of experimentation' became more and more centrally generated, with schemes that involved encouragement, funding, evaluation, and dissemination. Given the diversity of institutional cultures and structures, however, lessons were also being rapidly learned at national level. Centrally *generated* schemes did not mean central *control* or institutional *uniformity* in implementation. Enterprise in Higher Education was initially directed at raising consciousness amongst students and staff of a narrow version of 'enterprise'. Though much of the implementation had implications for skills acquisition and understanding of the needs of the workplace, in practice a central cascading model encountered strong counter-reactions. EHE meant, for example, student profiling and employer involvement, but it also meant a range of initiatives and developments – often 'innovations' – originating with individuals or the teaching-oriented units of institutions. These included curricular developments such as materials production and assessment experiments, but also, for instance, addressed equal opportunities issues and promoted student-centred teaching initiatives. In many institutions new teaching-focused structures and staff development activities were outcomes of EHE policies. EHE funding enabled ideas for change to be put into practice, with available finance, staff time, encouragement, networking with like-minded colleagues, and expert support.

Although central policy on innovation could (like all policy making) be transformed in the implementation process, 'directed innovation' was a defined framework within which initiatives were possible. The framework was more or less rigid and matched to varied extents the purposes of innovators at different levels in institutions. National programmes to promote the use of communication and information technologies (including the Teaching and Learning Technology Programme) were more prescriptive as to projects and outcomes. The Fund for the Development of Teaching and Learning, although largely subject-specific and with defined aims, opened up extensive possibilities for initiatives by teaching staff. Bidding processes, such as those regularly undertaken by the Department for Education and Employment, were policy-driven and related to specified areas (such as work-based learning or key skills) but these also offered scope for individuals, groups or units to pursue aims relating to their curricular and teaching experience. All such centrally promoted schemes provided important incentives to approach teaching and learning issues on a collaborative basis within and across institutions. Despite its initial narrow motivation EHE profoundly influenced the trend towards directed, policy-driven frameworks for innovation. In one sense the trend closed off or discouraged individually determined directions, but it also offered opportunities for interpretation and flexible initiatives within the defined boundaries. Outside those boundaries other kinds of innovation did not die, but they became more sporadic and difficult to sustain.

'Guided innovation' was being not so much transformed as merged into forms of 'directed innovation'. Incentives to innovate came from national government and its agencies and from institutional management, but remained a possibility for individuals facing, in most institutions, profound changes in the teaching/learning dialogue, influenced by increased student numbers, more diverse student constituencies and moves in such directions as skills-oriented curricula and learning outcomes. The reward system for staff was increasingly shaped in the 1980s and 1990s by an emphasis on research, with promotion more strongly tied to publication and research funding. There was throughout the period resistance among both teachers and students to moving away from traditional teaching methods and to using 'instructional media' (Moore and Hunt, 1980) and student-centred approaches. For example, many students in what the Americans tended to call 'the me generation' or under mounting financial pressures preferred what they saw as examination-oriented methods (Silver and Silver, 1997, ch. 7). For many staff, however, it was clear that the old forms of lecture and seminar were not working. Student constituencies were becoming greatly more diverse, expectations were different, the burden of assessment was becoming unbearable, the new and pervasive shapes of modular and semesterised courses were changing teaching and learning styles and rhythms, and the outcomes of higher education were being questioned by employers and the professions. The results at institutional or professional levels ranged from the adoption of problem-based learning for many programmes preparing the professions relating to medicine, to the adoption of new staff development programmes for new and existing teachers.

4. Typologies and Meanings

In the period from the early 1960s and 1990s it is possible to place innovations in, or directly influencing, teaching and learning in a rough typology:

- *Individual and group innovations*: classroom and course related, a direct response to student needs and professional concerns (student-led seminars, laboratory simulations).

- *Disciplinary initiatives*: sponsored or encouraged by subject associations or by professional or profession-related bodies such as the General Medical Council; informal collaboration amongst subject specialists across institutions.
- *Innovations responding to the educational media*: taking advantage of new technologies and acquiring or developing associated materials (software, e-mail, open or resource-based learning materials).
- *Curriculum-prompted innovations*: to meet the needs of modular and semesterised structures (including new assessment procedures) and in response to the changing content of fields of study and inter-disciplinary developments.
- *Institutional initiatives*: including policy decisions of many kinds (regarding information technology, work-based or resource-based learning) and staff development processes; new structures, including educational development units and similar bodies, teaching and learning committees, and the appointment of senior management to oversee the developments (pro-vice chancellors, deans).
- *Systemic initiatives*: including government creation of new and in various ways different kinds of institution (Open University, the 'green fields' universities), the funding of system-wide change (Enterprise in Higher Education, work- and skills-related developments); national agency schemes to extend the use of computers and educational technology; national pressure groups (Royal Society, Higher Education for Capability, Open Learning Foundation).
- *Systemic by-products*: resulting within higher education institutions from system-wide policies and practices (Teaching Quality Assessment, changes in student funding).

Any such typology, of course, acutely raises the question of whether there is any real meaning to the concept of 'innovation'. An analysis such as one based on individual, guided, and directed innovation overarches a more detailed typology, but both approaches point outwards from the innovative event to the nature of structural and policy changes in institutions and the system. The meanings ascribed to 'innovation' have to be sought in national policy and politics and in institutional responses and aspirations. A different typology might be constructed based on conceptions of 'conservative' and 'radical' innovation, polishing or undermining the status quo (Ayscough, 1976; Berg and Ostergren, 1979; Boyle, 1978; Levine, 1980). The pursuit in depth of any typology would also mean considering the profound differences of approach to teaching and learning, and to forms of innovation in general, amongst institutions with different histories, statuses, and ambitions.

Different types of innovation call for different requirements in relation to financial and moral support, and different opportunities for access to both in different types of institution. These institutional differences also account for the position of innovation in teaching and learning in competition with traditional or innovative approaches to the promotion and conduct of research and other commitments of academics within their institutional and disciplinary cultures. The study of innovation in teaching and learning is a study of interactions, attitudes, institutional policies and practices, national contexts, and the consensual and confrontational characteristics of all of them. 'To stimulate innovation' means asking not only 'what kind of innovation', but also 'whose innovation, in whose interests and in what, if any, policy contexts?

5. Note

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